2.3. VOLATILITY

Volatility is an extremely important risk factor. I measure volatility risk using the VIX index, which represents equity market volatility.[6] Here's a table correlating changes in VIX with stock and bonds returns on a monthly frequency basis from March 1986 to December 2011:

	VIX Changes	**Stock Returns**	**Bond Returns**
VIX Changes	1.00	−0.39	0.12
Stock Returns	−0.39	1.00	−0.01
Bond Returns	0.12	−0.01	1.00

The correlation between VIX changes and stock returns is –39%, so stocks do badly when volatility is rising. The negative relation between volatility and returns is called the *leverage effect*.[7] When stock returns drop, the financial leverage of firms increases since debt is approximately constant while the market value of equity has fallen. This makes equities riskier and increases their volatilities. There is another channel where high volatilities lead to low stock returns: an increase in volatility raises the required return on equity demanded by investors, also leading to a decline in stock prices. This second channel is a time-varying risk premium story and is the one that the basic CAPM advocates: as market volatility increases, discount rates increase and stock prices must decline today so that future stock returns can be high.[8]

Bonds offer some but not much respite during periods of high volatility, as the correlation between bond returns and VIX changes is only 0.12. Thus, bonds are not always a safe haven when volatility shocks hit. In 2008 and 2009, volatility was one of the main factors causing many risky assets to fall simultaneously. During this period, risk-free bonds did very well. But during the economic turbulence of the late 1970s and early 1980s, bonds did terribly, as did equities (see chapter 9). Volatility as measured by VIX can also capture uncertainty—in the sense that investors did not know the policy responses that government would take during the financial crisis, whether markets would continue functioning, or whether their own models were the correct ones. Recent research posits uncertainty risk itself as a separate factor from volatility risk, but uncertainty risk and volatility risk are highly correlated.[9]

[6] The VIX index is a measure of option volatilities on the S&P 500 constructed by the Chicago Board Options Exchange. The VIX index captures a variety of risks related to higher movements, including volatility itself, but also jump risk and skewness risk. But the main components captured in the VIX index are volatility and the volatility risk premium.

[7] A term coined by Fischer Black (1976).

[8] See equation (6.2) in chapter 6. For evidence of this time-varying risk premium channel, see Bekaert and Wu (2000).

[9] See, for example, Anderson, Ghysels, and Juergens (2009) for stocks and Ulrich (2011) for bonds.